| Module | Content | Hours | COs |
|---|---|---|---|
| 0 | **Prerequisite:** Basics Of Programming Language [ Syntax and Semantics], Computer Architecture. | 0 | |
| I | **Introduction to Regular Languages**<br>Languages: Alphabets and Strings. Regular Languages: Regular Expressions, Regular Languages, Regular Grammars, RL and LL grammars, Closure properties<br>**Self Learning:** Practice exercise on Regular Expressions. Identify the tools also. | 8 | CO1 |
| II | **Finite Automata and machines**<br>Finite Automata: FA as language acceptor or verifier, NFA ( with and without ε) , DFA, RE to NFA, NFA to DFA, Reduced DFA , NFA-DFA equivalence, FA to RE.<br>Finite State Machines with output : Moore and Mealy machines. Moore and Mealy M/C conversion. Limitations of FA.<br>**Self Learning :** Practice exercise on FA and NFA | 8 | CO2 |
| III | **Context Free Grammars**<br>Context Free Languages: CFG, Leftmost and Rightmost derivations, Ambiguity, Simplification and Normalization (CNF & GNF) and Chomsky Hierarchy ( Types 0 to 3)<br>**Self Learning :** Practice numerical or exercise on CFG | 8 | CO3 |
| IV | **Push Down Automata**<br>Push Down Automata: Deterministic (single stack) PDA, Equivalence between PDA and CFG. Power and Limitations of PDA.<br>**Self Learning :** List the examples of PDA. | 7 | CO4 |
| V | **Turing Machine**<br>Turing Machine: Deterministic TM, Multi-track and Multi-tape TMs, Concept of UTM and idea of system program. Issue and concept of Halting Problem<br>**Self Learning :** Practice numerical of TM. | 8 | CO5 |
| VI | **Application s of Automata**<br>Applications of FA, CFG, PDA & TM. Introduction to Compiler & Its phases.<br>**Self Learning :** Case study on any one compiler. | 6 | CO6 |
| | **Total:** | 45 | |

## Evaluation and Assessment Scheme:

### A. Internal Assessment Examination (IAE):

Assessment consists of two class tests, each 20 marks. The IAE 1 will cover any three Course Outcomes (COs) and IAE 2 will cover the remaining three Course Outcomes (COs). Each test will have a duration of one hour.

### B. End Semester Theory Examination (ESE):

End Semester exam of 60 Marks will be conducted based on entire syllabus.